

Solar Photovoltaic Remote Monitoring System Using IOT

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Abstract—As technology is advancing, cost of renewable energy equipments is decreasing which has resulted in a massive increase in solar photovoltaic installations. Most of these installations act as auxiliary power source. A majority of these are installed in inaccessible locations – as close as a rooftop to as far away as a desert. Hence they require sophisticated systems for remote monitoring of these installations using wide area networks. In this paper we will discuss a low cost IOT based embedded Solar PV Monitoring system which will make use of GPRS module and a low cost microcontroller to send the data measured at the production end on the internet, which can then be accessed anywhere on the globe. This will provide us real time information of the installation which will help us in its maintenance, fault detection and will give us a record of all the data at fixed intervals.

Keywords —Microcontroller, GPRS module, remote monitoring, solar photovoltaic, IOT

I. INTRODUCTION

Internet of things or IOT [1-2] is a revolution in the world of electronics. The idea is to connect all sensors and devices on a common network i.e. internet through wired or wireless means so that the user can access the data and control the devices from anywhere around the globe with an internet connection. Along with this, the aim is to automate the process using pre-defined logic to reduce human intervention as much as possible. It has gained a lot of attention in recent years due to its diversified usage in the fields of consumer electronics, home automation, health care, smart car, smart city and security purposes [3].

Solar energy is clean, abundant and an easily harnessed form of energy. Solar energy, although unreliable, is becoming more and more popular with advancement in technology and decreasing cost. With modern monitoring and control systems, these are becoming increasingly reliable sources of energy and in some years, might even replace conventional sources completely.

Power generated from Solar PV installations is susceptible to changes due to changes in solar irradiance, temperature, weather conditions and many other factors.

Monitoring of such installations is hence essential [4]. IOT based wireless system is chosen in order to avoid hazards [5] associated with wired systems, while keeping in mind the needs of near future, where every device will be smart, automated and connected via internet.

The experimental setup of proposed conceptual system consists of solar panels, voltage transducer, Hall Effect current sensor, temperature sensor LM35, SIM900A GPRS module and Arduino Uno microcontroller. Programming codes are developed on MATLAB and visualization of data is done on a website designed.

This system is capable of measuring voltage and current output, ambient temperature, comparing the data with previous databases and reference values and raising an alarm via a message in case of an anomaly, and sending the data over to a website which can be accessed via internet anywhere.

II. OVERALL SYSTEM DESIGN

The system described in this paper is capable of measuring the values of Solar PV voltage and current and temperature and sending them over on mobile networks to the internet for data logging and review by the user. In case of a deviation from normal specified values of current, voltage or temperature, the system is also capable of alerting the user via an SMS.

The Solar PV voltage is sensed by voltage divider circuit. Current is sensed using Hall Effect Current Sensor. The temperature is sensed using LM35 temperature sensor. The data obtained is processed using Arduino Uno microcontroller and sent as serial input to the SIM 900 GSM module. The GSM module sends the data over to the internet. A dedicated computer system is set up to store the data obtained for future reference and review. This data now can be viewed anywhere on the globe using an internet connection on an IP set up by the GSM module. Schematic diagram of the system is shown in figure 1 below.

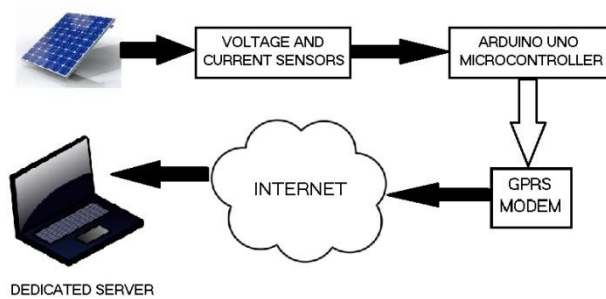


Figure 1: Schematic Diagram of Solar Photovoltaic Remote Monitoring System Using IOT

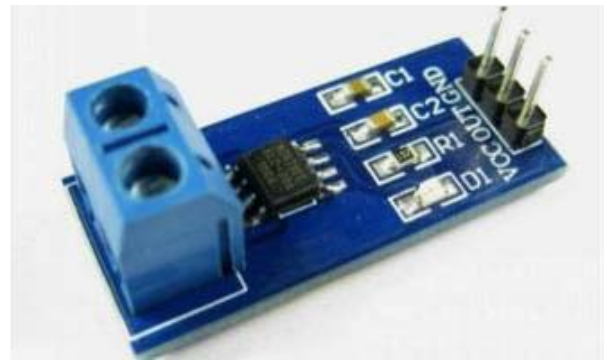


Figure 3: Hall Effect Current Sensor

III. COMPONENTS OF THE SOLAR PV REMOTE MONITORING SYSTEM USING IOT

A. Solar PV Installation

An installation of eight 125W Solar Panels of Emeral Energy Solutions Pvt. Ltd., comprising a total of 1kW system, shown in figure 2, is used to provide the required output to be measured. These have a maximum power output of 125 W, maximum voltage of 18.26 V and maximum current of 6.86 A.



Figure 2: 1 KW Solar Panel Installation comprising of 8 panels of 125 W each, installed on the roof-top of Renewable Energy Lab, MANIT, Bhopal.

B. Voltage and Current Transducers

The Solar PV voltage is sensed using a voltage divider circuit.

Current is sensed using Hall Effect Current Sensor. Hall Effect is the generation of potential difference due to a current carrying conductor in a perpendicular magnetic field [7]. Hall Effect current sensors work on this principle.

C. Temperature Sensor

The LM35 Integrated Circuit Temperature Sensor is used to measure the ambient temperature. It gives very accurate readings in Degree Celsius for over a range of 0-100°C [8] while drawing very low current of about 60 micro Amps.

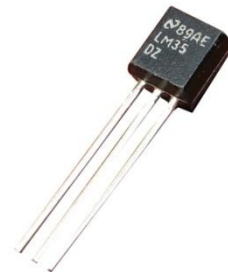


Figure 4: Temperature Sensor LM35

D. Low Cost Microcontroller

In this proposed conceptual system, the Arduino board acts as a micro-processor which processes the incoming data from various sensors. This board is selected because it reduces the requirement of various components like constant DC voltage regulator, burner hardware and does not requires a separate software to convert the code into hex file and burn it in the microprocessor. All work is done by Arduino IDE and is burnt in Arduino board via a USB chord.

Arduino Uno is a microcontroller board based on ATmega328P processor [9]. It can be coded as required using Arduino IDE in C/C++ language and uploaded directly to the board. This board has a plethora of applications in the field of embedded electronics. This is where it is decided whether the values obtained by various sensors are within limit depending upon the logic code given to Arduino board.

It processes the incoming data from various sensors and sends them to the GSM module via serial communication protocol along with various AT commands [11] necessary for the operation of GSM module.



Figure 5: Arduino UNO Microcontroller Board

E. GSM Module

SIM 900A GSM GPRS Module is used in this proposed conceptual system. It is a very reliable wireless module [10]. Its attributes are its small size, low cost and low power consumption. It delivers GSM/GPRS 900/1800 MHz performance for voice, SMS, Data and Fax.

In this system, it acquires the data to be sent to the user from Arduino board via serial communication along with the necessary AT commands and sends the data on the internet on a self created IP address. The user can enter this IP address on his system and get real time state condition of the Solar PV installation.

GSM module is chosen over Zigbee module because Zigbee technology [11] is proven inefficient in large scale because it can't face up huge distances.

Figure 6: SIM 900A GSM Module

IV. WORKING OF SOLAR PV REMOTE MONITORING SYSTEM USING IOT

The major role of this system is to send the data on the internet through GPRS enabled modem [12]. GPRS services of the modem are initiated using the AT commands along with the IP address and corresponding port. Values from various sensors are taken by the Arduino board and sent to the modem as a single string which should be of length less than 150 bytes [3]. Each value can be separated by commas or colons. The data is sent in ASCII format. The flow chart of working of the system is shown in figure 7.



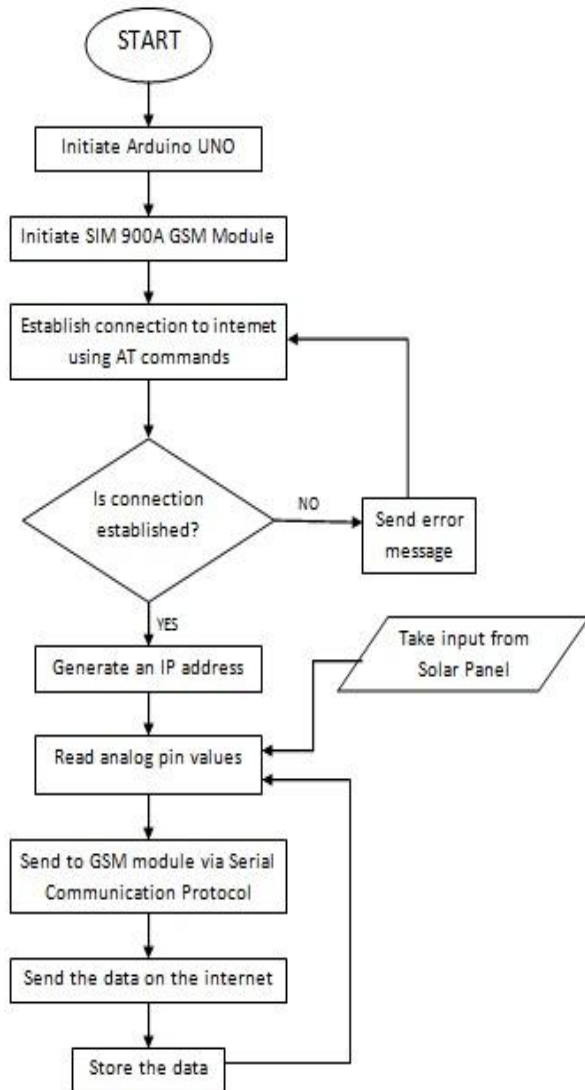


Figure 7: Flow Chart of Working of the Solar PV Remote Monitoring System Using IOT

The data is sent to a target server PC through the static IP. A data logging software will be running on the target PC and the data received from the GPRS module in ASCII format will be stored in a text file.

A standalone web hosting service will also be running on the target PC which would enable the user to access the data and monitor the Solar PV installation anywhere on the globe using internet connection.

V. CONCLUSION

As the conventional sources of electricity generation are depleting, mankind is in need of renewable sources such as solar and wind energy to sustain itself. The clean and abundant solar energy is a good alternative as a source of energy with the only problems of cost of harnessing solar energy, and its variable nature. With technological advancements, cost of devices is decreasing with a rapid rate. Hence all we need is a good, up-to-date monitoring system which can perform major tasks automatically without human intervention and can provide data to the user whenever and wherever needed.

To cope up with rapidly changing technology, IOT is the best solution for monitoring of solar installations. IOT based remote monitoring of the Solar PV installation will also save energy [14] and man-labour. Because of the use of IOT in this proposed system, there is a large scope for future work. can add modern devices and sensors without the fear of compatibility. Flexibility of this system is its uniqueness. Adding more sensors, it can measure AC voltage and current output, power consumption of load, solar irradiance and corresponding output of the solar panels and a lot more.

Further extremely useful for wireless IOT based monitoring and control to improve conventional solar based electrical vehicle system [15] for converter design and the adoption of suitable Maximum Power Point Tracking (MPPT) techniques [16].

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